

Quantum Mechanics III

Homework 3

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Topic

Quantum Channels and Kraus Representations

Submission

Submit one PDF. Show the derivation, identify every approximation, and include a short consistency check. The maximum file size is 25 MB.

Problems

1. Prove the first displayed relation used in Quantum Channels and Kraus Representations. State all assumptions and conventions.

$$\mathcal{E}(\rho) = \sum_a K_a \rho K_a^\dagger$$

2. Calculate a nontrivial case using the second relation. Carry units and uncertainty where applicable.

$$\sum_a K_a^\dagger K_a = I$$

3. Interpret the third relation in two limiting regimes and explain the physical meaning of the result.

$$J_{\mathcal{E}} = (I \otimes \mathcal{E})(|\Phi\rangle\langle\Phi|)$$

Show enough intermediate work that another student can reproduce the calculation.